

Notes A Functorial Excursion Between Algebraic Geometry and Linear Logic

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1 Basics

Let us start by recalling basic principles of algebraic geometry. The intuition is that a space X comes along with a topological space (X, Σ_X) such that every open set U is equipped with a set $\mathcal{O}_X(U)$ of local operations called *regular functions*. Such regular functions are typically defined as polynomials with division and each $\mathcal{O}_X(U)$ defines a commutative ring. In functorial terms, every space gives rise to a ringed space, that is, a topological space (X, Σ_X) equipped with a contravariant functor $\mathcal{O}_X : \Sigma_X^{op} \rightarrow \mathbf{Ring}$ from the set of opens Σ_X^{op} with the reversed inclusion order. The functorial action of $\mathcal{O}_X$ describes the restriction functions

$$\mathcal{O}_X(V \subseteq U) : \mathcal{O}_X(U) \rightarrow \mathcal{O}_X(V)$$

which takes an operation $f \in \mathcal{O}_X(U)$ defined on an open set U and restrict it to an operation

$$f|_V := \mathcal{O}_X(V \subseteq U)(f)$$

on an open subset $V \subseteq U$. The functorial understanding of the concept of a space is supported by the observation that every commutative ring R defines a ringed space:

$$\mathrm{Spec} R = (\mathrm{Spec} R, \mathcal{O}_{\mathrm{Spec} R})$$

on the set of prime ideals $\mathfrak{p}$ of the commutative ring R . Such a ringed space is called the *affine scheme* associated to R . Every space X of the theory, called a *scheme*, is a patchwork of affine schemes $\text{Spec } R_i$ glued together using the language of sheaf theory.

Every scheme X comes equipped with a symmetric monoidal category $\mathbf{qcMod}_X$ of quasicoherent sheaf of $\mathcal{O}_X$ -modules. By a presheaf of $\mathcal{O}_X$ -modules M of a ringed space $(X, \mathcal{O}_X)$, we mean a contravariant functor

$$M : \Omega_X^{op} \rightarrow \mathbf{Ab}$$

where $\mathbf{Ab}$ is the category of Abelian groups such that each Abelian group $M(U)$ defines a $\mathcal{O}_X(U)$ -module. The Serre-Swann theorem states a vector bundle on a suitable ringed space $(X, \mathcal{O}_X)$ can be equivalently encoded as its sheaf of sections that defines a locally free and projective sheaf of $\mathcal{O}_X$ -modules. A sheaf of $\mathcal{O}_X$ is called quasicoherent when it is locally presentable, that is, it is locally the cokernel of a morphism of free modules. The category $\mathbf{qcMod}_X$ extends the category of vector bundles to define an Abelian category where kernels and direct images can be computed.

The idea is to associate a model of intuitionistic linear logic, where formulas are understood as generalised vector bundles and proofs are vector fields by building an appropriate linear-non-linear adjunction. To associate a model of linear logic with each scheme X , we take the category $\mathbf{PshMod}_X$ of presheaves of $\mathcal{O}_X$ -modules and their homomorphisms as the underlying symmetric monoidal closed category with tensor product $\otimes_X$, the structural sheaf $\mathcal{O}_X$ as unit, and $-\circ_X$ for internal homs. For the cartesian part of a linear-non-linear adjunction, we take the category $\mathbf{PshCoAlg}_X$ of cocommutative comonoids (or commutative coalgebras) in $\mathbf{PshMod}_X$. It is convenient to the approach based on the functor of points (see [EH00, I.4]). Spec defines a contravariant functor $\text{Spec} : \mathbf{Ring}^{op} \rightarrow \mathbf{Scheme}$ that induces the *nerve functor*

$$\text{nerve} : \mathbf{Scheme} \rightarrow [\mathbf{Ring}, \mathbf{Set}]$$

associating to every commutative ring R the set of R -points in a scheme X defined as $\mathbf{Scheme}(\text{Spec } R, X)$. The nerve functor is fully faithful and it allows one to see every scheme X as a covariant presheaf

$$\text{nerve}(X) : \mathbf{Ring} \rightarrow \mathbf{Set}$$

So we move to schemes to $\mathbf{Ring}$ -spaces and associate a model of intuitionistic linear logic with each $\mathbf{Ring}$ -space X , where formulas are interpreted in $\mathbf{PshMod}_X$ as presheaves of $\mathcal{O}_X$ -module understood as generalised vector bundles on the $\mathbf{Ring}$ -space. The linear-non-linear adjunction (and therefore the exponential modality) is constructed by the Sweedler construction.

2 On the category of rings

Let $(\mathcal{A}, \otimes, \mathbb{1})$ be a symmetric monoidal category with reflexive coequalisers preserved by tensor products in every coordinate. For example, the category $\mathbf{Ab}$

of Abelian groups and homomorphisms. A ring is a monoid object (R, m, e) in $(\mathcal{A}, \otimes, \mathbb{1})$. In other words, a ring is a triple such that (R, m, e) consisting of an object R and morphisms $m : R \otimes R \rightarrow R$ and $e : \mathbb{1} \rightarrow R$ such that the following diagrams commute:

$$\begin{array}{ccc} R \otimes R \otimes R & \xrightarrow{m \otimes R} & R \otimes R \\ R \otimes m \downarrow & & \downarrow m \\ R \otimes R & \xrightarrow{m} & R \end{array} \quad \begin{array}{ccc} R \otimes R & \xrightarrow{m} & R \\ m \downarrow & \nearrow & \downarrow e \otimes R \\ R & \xrightarrow{R \otimes e} & R \otimes R \end{array}$$

A ring (R, m, e) is commutative is the following triangle commutes:

$$\begin{array}{ccc} R \otimes R & \xrightarrow{\gamma_{R,R}} & R \otimes R \\ & \searrow m & \swarrow m \\ & R & \end{array}$$

A *ring morphism* $u : (R, m_R, e_R) \rightarrow (S, m_S, e_S)$ is a morphism $u : R \rightarrow S$ in the underlying category $\mathcal{A}$ making the following diagrams commute:

$$\begin{array}{ccc} R \otimes R & \xrightarrow{u \otimes u} & S \otimes S \\ m_S \downarrow & & \downarrow m_S \\ R & \xrightarrow{u} & S \end{array} \quad \begin{array}{ccc} & \mathbb{1} & \\ e_R \swarrow & & \searrow e_S \\ R & \xrightarrow{u} & S \end{array}$$

The category **Ring**($\mathcal{A}$) of rings in $\mathcal{A}$ defined by the tensor product $R, S \mapsto R \otimes S$ in $\mathcal{A}$.

3 On the category of R -modules and modules

Given a ring R in a symmetric monoidal category $\mathcal{A}$, an R -*module* is a pair (M, act) where M is an object in $\mathcal{A}$ along with a morphism $\text{act} : R \otimes M \rightarrow M$ in $\mathcal{A}$ such that:

$$\begin{array}{ccc} R \otimes R \otimes M & \xrightarrow{m_R \otimes M} & R \otimes M \\ R \otimes \text{act} \downarrow & & \downarrow \text{act} \\ R \otimes M & \xrightarrow{\text{act}} & M \end{array} \quad \begin{array}{ccc} & R \otimes M & \\ e_{R \otimes M} \nearrow & & \searrow \text{act} \\ M & \xlongequal{\quad} & M \end{array}$$

Equivalently, an R -module is an Eilenberg-Moore algebra for the monad $A \mapsto R \otimes A$. An R -module homomorphism $f : M \rightarrow N$ between two modules is a map f in $\mathcal{A}$ such that the following diagram commutes:

$$\begin{array}{ccc} R \otimes M & \xrightarrow{R \otimes f} & R \otimes N \\ \text{act}_M \downarrow & & \downarrow \text{act}_N \\ M & \xrightarrow{f} & N \end{array}$$

$\mathbf{Mod}_R$ is the category of R -modules and their morphisms.

A *module* is a pair (R, M) where R is a ring and M is a module. A *module homomorphism* $(u, f) : (R, M) \rightarrow (S, N)$ is a ring morphism $u : R \rightarrow S$ and a morphism $f : M \rightarrow N$ such that

$$\begin{array}{ccc} R \otimes M & \xrightarrow{u \otimes f} & S \otimes N \\ \text{act}_M \downarrow & & \downarrow \text{act}_N \\ M & \xrightarrow{f} & N \end{array}$$

$\mathbf{Mod}$ is the category of modules and their morphisms.

There is a functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ transporting every module (R, M) to the underlying ring R and every morphism (u, f) to u . We will write

$$u : R \rightarrow S \models f : M \rightarrow N$$

for a module morphism $(u, f) : (R, M) \rightarrow (S, N)$. The intuition behind the notation is that every ring homomorphism $u : R \rightarrow S$ induces a “fibre” consisting of all module morphisms $(u, f) : (R, M) \rightarrow (S, N)$ living above and refining the ring morphism $u : R \rightarrow S$. The fibre of πR for a commutative ring is the category $\mathbf{Mod}_R$. The functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ defines a Grothendieck fibration: every pair consisting of a ring morphism $u : R \rightarrow S$ and of an S -module (N, act_N) induces an R -module noted

$$\mathbf{res}_u N = (N, \text{act}'_N)$$

with the action morphism act'_N defined as the composite:

$$R \otimes N \xrightarrow{u \otimes N} S \otimes N \xrightarrow{\text{act}_N} N$$

Moreover, the S -module with a module morphism:

$$u : R \rightarrow S \models \text{id}_N : \mathbf{res}_u N \rightarrow N$$

which is weakly Cartesian in the sense that every module morphism

$$u : R \rightarrow S \models f : M \rightarrow N$$

factors uniquely as

$$M \xrightarrow{(id_R, h)} \mathbf{res}_u N \xrightarrow{(u, id_N)} N$$

for a morphism $h : M \rightarrow \mathbf{res}_u N$ in the category $\mathbf{Mod}_R$. The unique solution h is equal to the original morphism $f : M \rightarrow N$ viewed as a morphism in $\mathbf{Mod}_R$. Given a ring morphism $u : R \rightarrow S$, the existence of a weakly Cartesian map for every S -module N ensures the existence of a functor

$$\mathbf{res}_u : \mathbf{Mod}_S \rightarrow \mathbf{Mod}_R$$

called restriction of scalar along u . Along with the fact that the composite of weakly Cartesian maps is weakly Cartesian we have that the forgetful functor $\pi : \mathbf{Mod} \rightarrow \mathbf{Ring}$ is a Grothendieck fibration.

Now we use that fact the category $\mathcal{A}$ has reflexive coequalisers, so one can show that $\mathbf{res}_u$ has a left adjoint

$$\mathbf{ext}_u : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_S$$

constructed as follows. Every ring homomorphism $u : R \rightarrow S$ induces an $R \otimes S$ -module notes as $R \otimes_u S$ defined as the reflexive coequaliser of the following diagram

$$\begin{array}{ccc} & \xrightarrow{m_R \otimes S} & \\ R \otimes R \otimes S & \xleftarrow{R \otimes e_R \otimes S} & R \otimes S \\ & \xrightarrow{(R \otimes m_S) \circ (R \otimes u \otimes S)} & \end{array}$$

computed in $\mathcal{A}$. Given three commutative rings R , S_1 and S_2 , we define the functor

$$\otimes_R : \mathbf{Mod}_{S_1 \otimes R} \times \mathbf{Mod}_{R \otimes S_2} \rightarrow \mathbf{Mod}_{S_1 \otimes S_2}$$

The two maps $\mathbf{act}_M : M \otimes R \rightarrow M$ and $\mathbf{act}_N : N \otimes R \rightarrow N$ are deduced by restriction of scalar from the $S_1 \otimes R$ -module structure of M and the $R \otimes S_2$ -module structure of N . The left adjunction functor $\mathbf{ext}_u$ is then defined the following way:

$$\mathbf{ext}_u : M \mapsto M \otimes_R (R \otimes_u S)$$

4 The category $\mathbf{Mod}^\ominus$ of modules and retromorphisms

Let $[M, N]$ denote the internal hom of M and N in $\mathcal{A}$. A *module retromorphism*

$$(u, f) : (S, N) \rightarrow (R, M)$$

is a pair (u, f) where $u : R \rightarrow S$ is a ring morphism and $f : N \rightarrow M$ is a morphism in $\mathcal{A}$ such that the following diagram commutes:

$$\begin{array}{ccccc} R \otimes M & \xleftarrow{R \otimes f} & R \otimes N & \xrightarrow{u \otimes N} & S \otimes N \\ \mathbf{act}_M \downarrow & & & & \downarrow \mathbf{act}_N \\ M & \xleftarrow{f} & & & N \end{array}$$

The category $\mathbf{Mod}^\ominus$ has ring modules as objects and module retromorphisms. There is a forgetful functor $\pi : \mathbf{Mod}^\ominus \rightarrow \mathbf{Ring}$ transporting every retromorphism to the underlying ring morphism and π is also a Grothendieck fibration. π is also a bifibration. Let $u : R \rightarrow S$ be a ring morphism, then functor defined by coextension of scalar along u

$$\mathbf{coext}_u : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_S$$

transport every R -module (M, act_M) to the S -module $[S, M]_u$ defined as the coreflexive equaliser of the below diagram:

$$[S, M] \begin{array}{c} \xrightarrow{[u \otimes S, M] \circ [m_S, M]} \\ \xleftarrow{[e_R \otimes M, M]} \\ \xrightarrow{[R \otimes S, \text{act}_M] \circ [R \otimes _, R \otimes _]} \end{array} [R \otimes S, M]$$

Note that $[S, M]_u$ provides an internal description of the set of morphisms $f : S \rightarrow M$ making the following diagram commute:

$$\begin{array}{ccc} R \otimes S & \xrightarrow{R \otimes f} & R \otimes M \\ u \otimes S \downarrow & & \downarrow \text{act}_M \\ S \otimes S & & \\ m_S \downarrow & & \\ S & \xrightarrow{f} & M \end{array}$$

or, equivalently, as the set of morphisms $f : \mathbf{res}_u S \rightarrow M$. Thus, every ring morphism $u : R \rightarrow S$ induces the following sequence of adjunctions:

$$\mathbf{Mod}_R \begin{array}{c} \xrightarrow{\mathbf{coext}_u} \\ \xleftarrow{\mathbf{res}_u} \\ \xrightarrow{\mathbf{ext}_u} \end{array} \mathbf{Mod}_S$$

5 The category $\mathbf{Mod}_R$ is symmetric monoidal closed

It is well known that the category $\mathbf{Mod}_R$ of modules over a ring R is symmetric monoidal closed. Its tensor product

$$\otimes_R : \mathbf{Mod}_R \times \mathbf{Mod}_R \rightarrow \mathbf{Mod}_R$$

transports every pair of R -modules (M, N) into the coreflexive equaliser of the following diagram calculated in $\mathcal{A}$:

$$M \otimes R \otimes N \begin{array}{c} \xrightarrow{\text{act}_M \otimes N} \\ \xleftarrow{M \otimes e_R \otimes N} \\ \xrightarrow{M \otimes \text{act}_N} \end{array} M \otimes N$$

In $\mathbf{Mod}_R$, the tensorial unit is the R itself seen as an R -module. The internal hom

$$[_, _]_R : \mathbf{Mod}_R^{op} \times \mathbf{Mod}_R \rightarrow \mathbf{Mod}_R$$

takes every pair of R -modules M and N to the coreflexive equaliser of the following diagram:

$$[M, N] \begin{array}{c} \xrightarrow{[\text{act}_M, N]} \\ \xleftarrow{[e_R \otimes M, N]} \\ \xrightarrow{[R \otimes M, \text{act}_N] \circ [R \otimes _, R \otimes _]} \end{array} [R \otimes M, N]$$

6 The category $\mathbf{Mod}$ as a symmetric monoidal ringed category

Let us consider the category $\mathbf{Mod} \times_{\mathbf{Ring}} \mathbf{Mod}$ defined by the pullback diagram as follows:

$$\begin{array}{ccc} \mathbf{Mod} \times_{\mathbf{Ring}} \mathbf{Mod} & \longrightarrow & \mathbf{Mod} \\ \downarrow & & \downarrow \pi \\ \mathbf{Mod} & \xrightarrow{\pi} & \mathbf{Ring} \end{array}$$

The fibrewise tensor product

$$\otimes_{\mathbf{Mod}} : \mathbf{Mod} \times_{\mathbf{Ring}} \mathbf{Mod} \rightarrow \mathbf{Mod}$$

transports every pair of modules (R, M) and (R, N) on the same monoid R to the R -module $(R, M \otimes_R N)$, so every pair of module homomorphisms:

$$\begin{array}{l} u : R \rightarrow S \models h_1 : M_1 \rightarrow N_1 \\ u : R \rightarrow S \models h_2 : M_2 \rightarrow N_2 \end{array}$$

above the same ring morphism $u : R \rightarrow S$ to the module homomorphism:

$$u : R \rightarrow S \models h_1 \otimes_u h_2 : M_1 \otimes_R M_2 \rightarrow N_1 \otimes_R N_2$$

where $h_1 \otimes_u h_2$ is a unique morphism making the following diagram commute:

$$\begin{array}{ccc} M_1 \otimes R \otimes M_2 & \xrightarrow{h_1 \otimes u \otimes h_2} & N_1 \otimes S \otimes N_2 \\ M_1 \otimes \text{act}_{M_2} \downarrow & \text{act}_{M_1} \otimes M_2 \downarrow & N_1 \otimes \text{act}_{N_2} \downarrow \text{act}_{N_1} \otimes N_2 \downarrow \\ M_1 \otimes M_2 & \xrightarrow{h_1 \otimes h_2} & N_1 \otimes N_2 \\ \text{quotient} \downarrow & & \downarrow \text{quotient} \\ M_1 \otimes_R M_2 & \xrightarrow{h_1 \otimes_u h_2} & N_1 \otimes_S N_2 \end{array}$$

The fibrewise unit $\mathbb{1}_{\mathbf{Mod}} : \mathbf{Ring} \rightarrow \mathbf{Mod}$ takes every commutative ring and converts it to itself viewed as a module. Categorically, one can understand that with the concept of a *ringed category*, which defined as a pair $(\mathcal{C}, \pi)$ where $\mathcal{C}$ is a category and $\pi : \mathcal{C} \rightarrow \mathbf{Ring}$ is a functor. The slice 2-category $\mathcal{C}/\mathbf{Ring}$ consists of ringed categories, fibrewise functions as 1-cells and natural transformations as 2-cells. The 2-category $\mathcal{C}/\mathbf{Ring}$ is Cartesian; the fibrewise product and the unit define a symmetric pseudomonoid structure on the ringed category $(\mathbf{Mod}, \pi)$ in $\mathcal{C}/\mathbf{Ring}$.

References

- [EH00] David Eisenbud and Joe Harris. *The geometry of schemes*. Springer, 2000.